

EngTrace Example Template

Blueprint

Template Definition

A steady-state CSTR is used for the consumption of **reactant_name** whose inlet molar flow rate is **F_A0**, and the desired outlet molar flow rate is **F_A**. If the rate of reaction is **neg_r_A**, what is the required reactor volume in liters?

Variable Generation

$$1.0 \leq F_{A0} \leq 5.0; \quad 0.01 \leq -r_A \leq 0.05; \quad 0.1 \leq F_A \leq F_{A0} - 0.1$$



Generated Instance

Question

A steady-state CSTR is used for the consumption of **Propylene** whose inlet molar flow rate is **2.50 mol/s**, and the desired outlet molar flow rate is **0.50 mol/s**. If the rate of reaction is **0.0200 mol/(L.s)**, what is the required reactor volume in liters?

Chain-of-Thought Solution

Step 1: State the CSTR Design Equation

$$V = (F_{A0} - F_A) / (-r_A)$$

Step 2: Substitute the given values.

$$V = (2.50 \text{ mol/s} - 0.50 \text{ mol/s}) / (0.0200 \text{ mol/(L.s)})$$

Step 3: Calculate the final volume.

$$V = 100.0 \text{ L}$$